

Homework 1 Solution

ECE 253 - Digital Image Processing
Prof. Truong Nguyen

Problem 1

Let $h(n) = \delta(n) - 2\delta(n - 1)$.

(a) Compute the DTFT $H(e^{j\omega})$

The DTFT of $h(n)$ is given by:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$$

Substituting $h(n) = \delta(n) - 2\delta(n - 1)$, we get:

$$H(e^{j\omega}) = 1 - 2e^{-j\omega}$$

(b) Compute the z-transform $H(z)$

The z-transform of $h(n)$ is given by:

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n}$$

Substituting $h(n) = \delta(n) - 2\delta(n - 1)$, we get:

$$H(z) = 1 - 2z^{-1}$$

(c) Pole-zero plot

The zeros are the solutions of $H(z) = 0$, i.e., $1 - 2z^{-1} = 0$, which gives $z = 2$. There is a pole at $z = 0$. The pole-zero plot shows a zero at $z = 2$ and a pole at the origin.

(d) Compute and sketch $|H(e^{j\omega})|$

Magnitude and Phase Response of $H(e^{j\omega})$

We have $h(n) = \delta(n) - 2\delta(n-1)$, and the DTFT is:

$$H(e^{j\omega}) = 1 - 2e^{-j\omega}$$

To compute the magnitude and phase response, follow these steps:

Step 1: Simplify $H(e^{j\omega})$ First, express $H(e^{j\omega})$ in a more convenient form:

$$H(e^{j\omega}) = 1 - 2(\cos(\omega) - j\sin(\omega))$$

This simplifies to:

$$H(e^{j\omega}) = (1 - 2\cos(\omega)) + j(2\sin(\omega))$$

We now have $H(e^{j\omega}) = a(\omega) + jb(\omega)$, where:

$$a(\omega) = 1 - 2\cos(\omega)$$

$$b(\omega) = 2\sin(\omega)$$

Step 2: Compute the Magnitude Response $|H(e^{j\omega})|$ The magnitude response is given by:

$$|H(e^{j\omega})| = \sqrt{a(\omega)^2 + b(\omega)^2}$$

Substituting $a(\omega) = 1 - 2\cos(\omega)$ and $b(\omega) = 2\sin(\omega)$:

$$|H(e^{j\omega})| = \sqrt{(1 - 2\cos(\omega))^2 + (2\sin(\omega))^2}$$

Expanding both terms:

$$|H(e^{j\omega})| = \sqrt{(1 - 4\cos(\omega) + 4\cos^2(\omega)) + (4\sin^2(\omega))}$$

Simplify using the identity $\sin^2(\omega) + \cos^2(\omega) = 1$:

$$|H(e^{j\omega})| = \sqrt{5 - 4\cos(\omega)}$$

Step 3: Compute the Phase Response $\angle H(e^{j\omega})$ The phase response is the angle of the complex number $H(e^{j\omega}) = a(\omega) + jb(\omega)$:

$$\angle H(e^{j\omega}) = \tan^{-1} \left(\frac{b(\omega)}{a(\omega)} \right)$$

Substitute $a(\omega) = 1 - 2\cos(\omega)$ and $b(\omega) = 2\sin(\omega)$:

$$\angle H(e^{j\omega}) = \tan^{-1} \left(\frac{2\sin(\omega)}{1 - 2\cos(\omega)} \right)$$

The magnitude response is:

$$|H(e^{j\omega})| = |1 - 2e^{-j\omega}| = \sqrt{1^2 + (-2)^2 - 2 \cdot 1 \cdot (-2) \cdot \cos(\omega)}$$

$$|H(e^{j\omega})| = \sqrt{5 + 4 \cos(\omega)}$$

This can be sketched as a function of ω .

(e) Compute and sketch $\angle H(e^{j\omega})$

The phase response is:

$$\angle H(e^{j\omega}) = \tan^{-1} \left(\frac{-2 \sin(\omega)}{1 - 2 \cos(\omega)} \right)$$

This can be sketched as a function of ω .

(f) Find the output $y(n)$ for $x(n) = \delta(n) + \delta(n - 1) + \delta(n - 2)$

The output $y(n)$ is the convolution of $h(n)$ with $x(n)$:

$$y(n) = h(n) * x(n) = (\delta(n) - 2\delta(n - 1)) * (\delta(n) + \delta(n - 1) + \delta(n - 2))$$

Performing the convolution, we get:

$$y(n) = \delta(n) + \delta(n - 1) - 2\delta(n - 1) - 2\delta(n - 2) - 2\delta(n - 3)$$

$$y(n) = \delta(n) - \delta(n - 1) - 2\delta(n - 2) - 2\delta(n - 3)$$

Problem 2

Let $h_1(n) = (-1)^n h(n)$.

(a) Find and sketch $h_1(n)$

Substituting $h(n) = \delta(n) - 2\delta(n-1)$, we get:

$$h_1(n) = (-1)^n(\delta(n) - 2\delta(n-1))$$

This gives:

$$h_1(n) = \delta(n) + 2\delta(n-1)$$

The sketch of $h_1(n)$ shows a positive impulse at $n = 0$ and another positive impulse at $n = 1$.

(b) Relationship between $H_1(z)$ and $H(z)$; $H_1(e^{j\omega})$ and $H(e^{j\omega})$

Using the time-shifting property of the z-transform, we get:

$$H_1(z) = H(-z)$$

For the DTFT:

$$H_1(e^{j\omega}) = H(e^{j(\omega+\pi)})$$

(c) Pole-zero plot of $H_1(z)$

The zeros of $H_1(z)$ are at $z = -2$, and the pole remains at $z = 0$. The pole-zero plot shows a zero at $z = -2$ and a pole at the origin.

(d) Compute and sketch $|H_1(e^{j\omega})|$ and $\angle H_1(e^{j\omega})$

We have $h_1(n) = (-1)^n h(n)$, and the DTFT is:

$$H_1(e^{j\omega}) = H(e^{j(\omega+\pi)})$$

This means:

$$H_1(e^{j\omega}) = 1 - 2e^{-j(\omega+\pi)}$$

Step 1: Simplify $H_1(e^{j\omega})$ We first expand the exponential:

$$H_1(e^{j\omega}) = 1 - 2e^{-j\omega}e^{-j\pi}$$

Since $e^{-j\pi} = -1$, we get:

$$H_1(e^{j\omega}) = 1 + 2e^{-j\omega}$$

Next, express it in terms of real and imaginary parts:

$$H_1(e^{j\omega}) = 1 + 2(\cos(\omega) - j\sin(\omega))$$

This simplifies to:

$$H_1(e^{j\omega}) = (1 + 2\cos(\omega)) + j(-2\sin(\omega))$$

Step 2: Compute the Magnitude Response $|H_1(e^{j\omega})|$ The magnitude response is:

$$|H_1(e^{j\omega})| = \sqrt{(1 + 2\cos(\omega))^2 + (-2\sin(\omega))^2}$$

Expanding the terms:

$$|H_1(e^{j\omega})| = \sqrt{(1 + 4\cos(\omega) + 4\cos^2(\omega)) + 4\sin^2(\omega)}$$

Simplify using $\sin^2(\omega) + \cos^2(\omega) = 1$:

$$|H_1(e^{j\omega})| = \sqrt{5 + 4\cos(\omega)}$$

Step 3: Compute the Phase Response $\angle H_1(e^{j\omega})$ The phase response is:

$$\angle H_1(e^{j\omega}) = \tan^{-1} \left(\frac{-2\sin(\omega)}{1 + 2\cos(\omega)} \right)$$

The magnitude response is:

$$|H_1(e^{j\omega})| = |1 + 2e^{-j\omega}|$$

The phase response is:

$$\angle H_1(e^{j\omega}) = \tan^{-1} \left(\frac{2\sin(\omega)}{1 + 2\cos(\omega)} \right)$$

These can be sketched as functions of ω .

****Problem 3:** $G_1(z) = 1 + \frac{1}{3}z^{-1}$ **

**** (a)** Find $G_2(z)$ such that $G(z) = G_1(z)G_2(z)$ is a linear-phase system. **

We are tasked with finding $G_2(z)$ such that the product $G(z) = G_1(z)G_2(z)$ forms a linear-phase system. For a linear-phase system, the zeros must be symmetrically located with respect to the unit circle.

Given $G_1(z) = 1 + \frac{1}{3}z^{-1}$, the system has a zero at $z = -\frac{1}{3}$. To achieve linear-phase, we need to add a reciprocal zero to make the system symmetric, which means adding a zero at $z = -3$.

Thus, $G_2(z)$ must have the form:

$$G_2(z) = 1 + \frac{1}{3}z$$

Now, the overall system $G(z)$ becomes:

$$G(z) = G_1(z)G_2(z) = \left(1 + \frac{1}{3}z^{-1}\right) \left(1 + \frac{1}{3}z\right)$$

**** (b)** Find $G(z)$. Compute its magnitude and phase responses. **

Now we compute $G(z)$. Multiply the two terms:

$$G(z) = \left(1 + \frac{1}{3}z^{-1}\right) \left(1 + \frac{1}{3}z\right)$$

Expanding this product:

$$G(z) = 1 + \frac{1}{3}z^{-1} + \frac{1}{3}z + \frac{1}{9}$$

Simplifying:

$$G(z) = 1 + \frac{1}{3}(z^{-1} + z) + \frac{1}{9}$$

Now, substitute $z = e^{j\omega}$ to find the frequency response $G(e^{j\omega})$:

$$G(e^{j\omega}) = 1 + \frac{1}{3}(e^{-j\omega} + e^{j\omega}) + \frac{1}{9}$$

Using the identity $e^{-j\omega} + e^{j\omega} = 2 \cos(\omega)$, we get:

$$G(e^{j\omega}) = 1 + \frac{2}{3} \cos(\omega) + \frac{1}{9}$$

Simplifying further:

$$G(e^{j\omega}) = \frac{19}{9} + \frac{2}{3} \cos(\omega)$$

Magnitude Response $|G(e^{j\omega})|$

Since $G(e^{j\omega})$ is real-valued, the magnitude response is simply the absolute value:

$$|G(e^{j\omega})| = \left| \frac{19}{9} + \frac{2}{3} \cos(\omega) \right|$$

This is the magnitude response of the system.

Phase Response $\angle G(e^{j\omega})$

Since $G(e^{j\omega})$ is real and non-negative, the phase response is:

$$\angle G(e^{j\omega}) = 0$$

for all values of ω .

**** (c) Plot the pole-zero of $G(z)$ on the unit circle ****

From the expression of $G(z)$, we see that the system has zeros at: - $z = -\frac{1}{3}$ (from $G_1(z)$),
- $z = -3$ (from $G_2(z)$).

The poles of the system are at $z = 0$, since there are no terms in the denominator. The zeros are symmetric with respect to the unit circle, ensuring linear phase.

The plot would show: - One zero inside the unit circle at $z = -\frac{1}{3}$, - One zero outside the unit circle at $z = -3$.

**** (d) Which linear-phase type is $G(z)$? ****

The system is **** Type II **** linear-phase, as the zeros are symmetrically located around the unit circle and the system has an **** odd **** number of zeros (2).

Problem 4: Type II Linear-Phase System $G_{II}(z)$

**** (a) Plot the pole-zero of $G_{II}(z)$ ****

We are tasked with adding zeros to make $G(z)$ a Type II linear-phase system. From the previous problem, we already have zeros at $z = -\frac{1}{3}$ and $z = -3$.

To make this a **Type II** system, we need to ensure that all zeros are symmetrically located about the unit circle, and we should have an **even number of zeros**. Therefore, we can add two zeros at reciprocal locations, for example, at: - $z = \frac{1}{3}$, - $z = 3$.

These added zeros ensure that the system remains symmetric, and it has an even number of zeros, making it Type II.

The pole-zero plot would now include: - Zeros at $z = -\frac{1}{3}, \frac{1}{3}, -3, 3$, - No poles (all are at the origin).

** (b) Find $G_{II}(z)$ **

The new transfer function $G_{II}(z)$ becomes:

$$G_{II}(z) = (1 + \frac{1}{3}z^{-1})(1 + \frac{1}{3}z)(1 - \frac{1}{3}z^{-1})(1 - \frac{1}{3}z)$$

Multiply the terms:

$$G_{II}(z) = \left(1 - \left(\frac{1}{3}\right)^2 z^{-2}\right) \left(1 - \left(\frac{1}{3}\right)^2 z^2\right)$$

Simplify:

$$G_{II}(z) = \left(1 - \frac{1}{9}z^{-2}\right) \left(1 - \frac{1}{9}z^2\right)$$

This is the transfer function of the Type II linear-phase system.

** (c) Compute and sketch $|G_{II}(e^{j\omega})|$ and $\angle G_{II}(e^{j\omega})$ **

Now, compute the DTFT $G_{II}(e^{j\omega})$:

$$G_{II}(e^{j\omega}) = \left(1 - \frac{1}{9}e^{-j2\omega}\right) \left(1 - \frac{1}{9}e^{j2\omega}\right)$$

Expanding:

$$G_{II}(e^{j\omega}) = 1 - \frac{1}{9}(e^{-j2\omega} + e^{j2\omega}) + \frac{1}{81}$$

Using $e^{-j2\omega} + e^{j2\omega} = 2 \cos(2\omega)$, we get:

$$G_{II}(e^{j\omega}) = 1 - \frac{2}{9} \cos(2\omega) + \frac{1}{81}$$

Simplifying:

$$G_{II}(e^{j\omega}) = \frac{80}{81} - \frac{2}{9} \cos(2\omega)$$

Magnitude Response $|G_{II}(e^{j\omega})|$

Since $G_{II}(e^{j\omega})$ is real, the magnitude response is simply:

$$|G_{II}(e^{j\omega})| = \left| \frac{80}{81} - \frac{2}{9} \cos(2\omega) \right|$$

Phase Response $\angle G_{II}(e^{j\omega})$

Since $G_{II}(e^{j\omega})$ is real and non-negative, the phase response is:

$$\angle G_{II}(e^{j\omega}) = 0$$

for all ω .

Problem 5

Let $h_2(n) = h(n) * h_1(n)$.

(a) Compute the convolution $h_2(n)$

Convolving $h(n)$ and $h_1(n)$ gives:

$$h_2(n) = (\delta(n) - 2\delta(n-1)) * (\delta(n) + 2\delta(n-1))$$

Performing the convolution, we get:

$$h_2(n) = \delta(n) - 4\delta(n-1) + 4\delta(n-2)$$

(b) Compute the z-transform $H_2(z)$

The z-transform is:

$$H_2(z) = H(z)H_1(z) = (1 - 2z^{-1})(1 + 2z^{-1})$$

Simplifying:

$$H_2(z) = 1 - 4z^{-2}$$

(c) Compute the DTFT $H_2(e^{j\omega})$

The DTFT is:

$$H_2(e^{j\omega}) = 1 - 4e^{-j2\omega}$$

This can be used to compute the magnitude and phase responses.

(d) Causality and stability

The system is causal because all terms of $h_2(n)$ involve non-negative powers of n . It is stable since the region of convergence includes the unit circle.

Problem 6

Given $x(n) = \cos(\omega_0 n)$ and $h(n)$:

(a) Compute the output $y(n)$

Using the frequency response, we compute $y(n)$ as:

$$y(n) = H(e^{j\omega_0}) \cdot \cos(\omega_0 n)$$

(b) Exact output when $\omega_0 = \frac{\pi}{3}$

Substitute $\omega_0 = \frac{\pi}{3}$ into the expression for $H(e^{j\omega_0})$.

(c) Filter type

Based on the magnitude response, the system acts as a high-pass filter.

(d) Sketch the output for the first 10 samples

Plot the output signal for $n = 0$ to $n = 9$.